

University of Konstanz
Department of Politics and Public Administration
Seminar: „Wikipedia Data Analysis”
Dr. Taehee Kim
Winter Semester 2024/2025
30.03.2025

Exploring gender bias in Wikipedia's Vital Articles

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A thematic and structural analysis of biographies

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1. INTRODUCTION

Since its creation in 2001, Wikipedia has become the world's largest online encyclopedia and has transformed the way millions of people access, create, and share knowledge. Today, it exists in over 300 languages, hosting more than 61 million articles (Wikimedia Foundation, n.d.). The English Wikipedia, its largest edition has nearly 7 million articles and is maintained by a community of volunteer editors contributing around 5 million monthly edits to its articles (Wikimedia Foundation, 2025). Consistently ranked among the most visited websites, Wikipedia plays a central role in shaping public understanding as one of the most influential sources of information on the internet (Greenstein and Zhou, 2012).

What makes Wikipedia unique is its open, collaborative structure: anyone can contribute, edit and discuss content. Articles are maintained and refined through community-driven processes of discussion, consensus and collaboration. According to one of its most fundamental principles, the policy of the 'Natural Point of View (NPOV)', Wikipedia aspires to exclusively host articles free from biases. However, despite honest efforts of inclusivity and neutrality, Wikipedia, just like any other encyclopedia, is not immune to the cultural and societal contexts in which it is produced. In fact, over the years, scholars have increasingly drawn attention to the presence of gender bias on Wikipedia, manifesting in terms of both representation as well as editorial participation. For instance, male contributors are estimated to account for up to 90% of the platform's editor community, which is said to reinforce inequalities in the representation of gendered topics (Falenska et al., 2021). Research suggests that women are underrepresented among Wikipedia's article subjects, with merely 19% of biographies corresponding to females (Tripodi, 2021). Further, these biographies tend to be less developed, shorter in content, as well as less likely to contain structured data or hyperlinks (Ford et al., 2023). Beyond the disparities in participation and representation, research has also highlighted qualitative differences in how women are portrayed. A well-known example of gendered framing in biographies is the Finkbeiner test (Finkbeiner, 2013), originally developed for science journalism, which highlights how articles about women often emphasize their gender and relationships, their family roles, and caregiving responsibilities, rather than their professional achievements. Similar patterns are also reflected in Wikipedia's biographies. Wagner et al. (2015) found that articles about women contain more family- and gender-related words and include more hyperlinks to articles about men than vice versa. In a follow-up study, Wagner et al. (2016) further demonstrated that women must meet a higher notability threshold to warrant a

Wikipedia biography, revealing another layer of gender-based asymmetry and inherent bias in selection criteria.

Much of the existing research so far has focused on large datasets of biographies irrespective of notability (Tripodi, 2021; Wagner et al., 2016), or on Wikipedia representations of notable individuals, identified specifically from external sources (Wagner et al., 2015; Callahan and Herring, 2011; Ford et al., 2023). Despite the attention given to gender bias in several areas, its persistence in Wikipedia’s highly curated and carefully maintained content – the Vital Articles list – remains an intriguing and underexplored topic. The Vital Articles list is a collection of English Wikipedia’s most important topics, including a selection of around 2,000 individuals who are considered globally significant. These articles are often subject to higher editorial scrutiny and stricter quality standards, while the list itself is actively maintained and updated. Therefore, these vital biographies offer a unique opportunity to explore whether said disparities persist even at the highest levels of visibility. As such, the following paper aims to provide and exploratory analysis of the presence of gender bias in the Vital Articles list of individuals on Wikipedia. Specifically, it examines whether biographies of women differ systematically from those of men in terms of article length, quality, structural complexity, and editorial activity. Further, the analysis extends to a thematic exploration of the content itself, by identifying recurring and dominant topics, as well as investigating the occurrence of family-related terminology in these biographies. The research question guiding this exploratory analysis is: Does gender bias, previously documented across Wikipedia biographies, also manifest the structure and content of biographies in Wikipedia’s Vital Articles list?

2. METHODOLOGY

This research focuses on a subset of Wikipedia’s ‘Vital Articles’ list, specifically the category of level-4 human biographies which included 1,997¹ articles at the time of data collection. This particular level was selected as it represents a broad, yet manageable set of biographies, providing a sufficient sample size to detect patterns of bias while remaining computationally feasible.

¹ Although the Wikipedia page initially listed 1,998 articles, a full re-extraction of the data on 26.03.2025 returned only 1,997. Since the extraction code remained unchanged and was carefully reviewed, the discrepancy likely resulted from the removal of one article from the list in the meantime.

2.1. Data collection and preprocessing

Data collection began by extracting the list of articles from the corresponding Wikipedia page via an HTTP request, followed by HTML parsing. For each entry, article title, relative URL, and Wikipedia-assigned quality ratings were extracted as the first step of data collection. The individual biography page titles, as well as relative URLs were identified by selecting `<a>` tags nested within `<li>` elements, while ensuring that the URLs referenced Wikipedia articles (i.e., relative links starting with `/wiki/`). Article rating was found within a `<span>` element with the class "noviewer", which contained either an `<a>` tag or, in some cases, a nested `<span>` element instead. To ensure all ratings were captured, both structures were checked and then extracted from the "title" attribute of these elements (e.g., "C-Class article"). Non-biographical pages, such as the Wikipedia *Main Page*, were identified and removed. After these adjustments, the dataset contained 1,997 unique biography pages.

Wikidata item IDs were then retrieved by querying Wikipedia's MediaWiki API, using the extracted relative links as identifiers. These requests were executed asynchronously to handle the volume of calls efficiently. One article did not return an ID through this method; after a thorough review of the extraction process, no issue with the code could be identified, and so the missing ID was manually added. Based on articles' Wikidata identifier (property P21), the gender of each individual was determined. Male and female categories were recorded, while all other gender-related classifications were grouped under the category 'other'. Upon inspection, it was found that entries under 'other' ($n = 26$) exclusively corresponded to groups (e.g. Brothers Grimm, Rothschild Family, ABBA) and were therefore excluded from analysis.

Additionally, several key article attributes were obtained using Wikipedia's API and page identifiers, to analyze differences between female and male biographies. The full text of each biography was retrieved and parsed to extract the main text corpus. It was then cleaned by removing non-relevant HTML elements (e.g. tables, reference markers, scripts), as well as sections like "References", "Notes" etc. to focus on the parts of the articles that would be relevant for subsequent text analyses. Internal Wikipedia hyperlinks were collected for each biography, whereby, *Main Page* links were excluded. Finally, full revision histories were retrieved to analyze editorial activity. Revision data was fetched, including details such as revision IDs, timestamps, users, and 'sha1' hash values of consecutive revisions, which helped identifying reverts, i.e. instances where edits restored previous versions of the articles. The

entirety of the collected data was initially stored in a dictionary format for easy access and was later organized into several thematic DataFrames for further analysis.

2.2. Analysis

Key page and editing characteristics

Part of the analysis involved examining various characteristics of Wikipedia biographies across gender categories, focusing on text and page attributes, as well as article quality. For this, several numerical variables related to text and page structure were computed. The number of sections in an article was determined by counting heading elements within the extracted HTML content. The number of internal hyperlinks for each page was counted and recorded, and article length was measured by the number of words in the fully cleaned and preprocessed text. Editorial activity was assessed by analyzing revision histories, capturing key metrics such as the total number of edits, the number of detected reverts, and number of unique editors per article. Further, revert rates were computed as the proportion of reverts relative to the number of revisions per page to account for differences in overall editing volume. To systematically compare these characteristics across gender categories, mean values were computed for each variable. A t-test was conducted to assess whether differences were statistically significant. Article quality was assessed through the predefined levels: "Start", "C-Class", "B-Class", "Good", and "Featured". The distribution of quality across genders was explored by computing the proportion of articles within each quality category, normalized by the total number of biographies per gender. Distribution plots were created to visualize these patterns.

Topic modeling and thematic analysis

To analyze thematic differences in biographies, Latent Dirichlet Allocation (LDA) topic modeling was applied to the textual content of articles. LDA is an unsupervised machine learning technique used for extracting underlying topics across documents, where each topic is characterized by a distribution of words. In the context of this analysis, LDA allows for the identification and direct comparison of dominant themes among female and male biographies. Prior to model training, the cleaned texts were preprocessed to ensure reliable analysis. This included tokenization, lowercasing, lemmatization, the removal of 'common words', punctuation, and numbers. Common words (e.g. "this", "and", "is") were removed using the NLTK English stopword list. In addition, I compiled a short list of costume words to exclude that can be broadly defined to belong in one of two categories: (1) commonly used, but for topic

extraction specifically meaningless words (e.g., “also”, “would”, “two”) and (2) time-related terms (e.g. “year”, “later”, “time”, and month names) that are often present in biographies but are not relevant for identifying gendered differences in the content. After preprocessing, biographies were grouped by gender and converted into a Bag-of-Words representation. Unique words were mapped to integer identifiers, and the corpus was transformed into a document-term matrix. LDA models were then trained on male and female biographies separately using eight topics, which was chosen based on exploratory testing to balance coherence and complexity. Finally, the most probable words representing each topic were extracted, their frequencies computed and visualized for the two genders separately.

Beyond topic modeling, the occurrence of words describing personal relationships, along with family-related terms in biographies was examined to explore potential gendered differences in how personal and familial contexts are discussed. To count occurrences and extract frequencies, a custom list of terms relating to family and relationships (e.g. “mother”, “husband”, “child”, “marry”, “widow”) was applied to the preprocessed biographies of the previous step. The word woman was intentionally excluded from the custom list to try and avoid skewed results, as it was expected to find some women’s rights activists among the Vital Articles. The results of this analysis were visualized using bar charts while differentiating between genders.

3. RESULTS

3.1. Key biography characteristics and editing activity

The dataset included a total of 1,971 biographies, of which 205 (10.40%) were about women and 1,766 (89.60%) about men. In terms of structural characteristics, there were no substantial differences between male and female biographies. *Table 1a* contains an overview of the average values and standard deviations of the number of words, sections, and hyperlinks across both genders. On average, articles about women were slightly longer and contained more hyperlinks, while the number of sections remained nearly identical. These differences were minimal, and not statistically significant, suggesting that overall article structure does not vary systematically by gender. While boxplots revealed moderate variability for both male and female biographies in article length and number of sections – with somewhat greater variation for the males - no substantial gender-based disparities could be identified in these characteristics. After filtering out values above the 99th percentile, the differences in average values remained approximately the same and were not statistically significant, therefore, those numbers are not included in this report.

Table 1a: Description of biographies - structural characteristics

	Words		Sections		Hyperlinks	
	Mean	SD	Mean	SD	Mean	SD
Female	4692.76	3164.19	25.98	11.38	1097.54	891.34
Male	4599.11	3693.85	26.20	12.79	980.36	781.87

Regarding editorial activity (*Table 1b*), biographies of women had on average, more revisions, reverts, and unique editors. The distribution plots of editorial activity (*Figure 1*) reveal that male biographies exhibit a much wider spread across all three aspects, with several values standing out as outliers. This suggests, that while there are quite a few male biographies that are heavily edited, or possibly contested, female biographies tend to receive a more consistent level of editorial attention. After filtering out outliers above the 99th percentile, which had skewed the male biography averages, the difference in editorial activity becomes even more pronounced, indicating an overall higher level of engagement with female biographies. These differences can be observed in *Table 1b*, where the means and standard deviations for the filtered data are shown (*in parentheses*). The results of the t-tests for the filtered dataset further confirm these patterns, revealing statistically significant differences across all three measures of editorial activity. Female biographies exhibit significantly higher levels of engagement in terms of revisions ($t = -2.494$, $p = 0.013$), reverts ($t = -2.115$, $p = 0.035$), and unique editors ($t = -2.671$, $p = 0.008$). The negative t-values indicate that, on average, female biographies receive more editorial attention than their male counterparts. However, this increased engagement does not translate into greater editorial disputes, as the mean per-page revert rates remain nearly identical at 15.4% for female biographies and 15.3% for male biographies.

Table 1b: Description of biographies – editing activity

	Revisions		Reverts		Editors	
	Mean	SD	Mean	SD	Mean	SD
Female	4197.98 (4045.36)	3392.13 (3036.62)	740.09 (740.09)	691.41 (691.41)	1750.64 (1726.84)	1171.53 (1123.65)
Male	3720.55 (3482.95)	3961.38 (3086.05)	682.30 (632.43)	862.80 (672.21)	1566.96 (1504.04)	1319.97 (1160.60)

* The numbers in parentheses represent the mean and standard deviation after filtering the dataset for outliers above the 99th percentile.

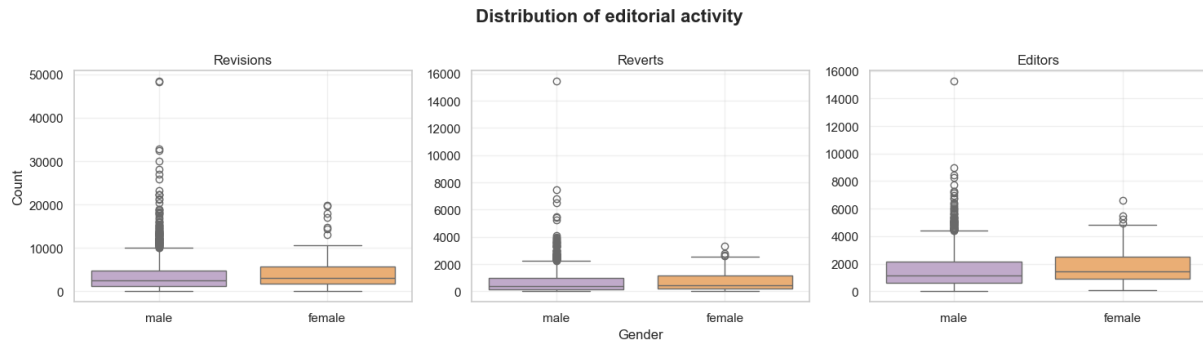


Figure 1: Boxplots depicting the distribution of editorial activity across genders

Shifting focus to article quality, the distribution of ratings (*Figure 2*) reveals notable differences between male and female biographies. While a higher proportion of male biographies fall into the lower-quality categories, especially into the C-Class article category (36.18% vs. 17.07%), female biographies are more frequently rated at higher quality levels. Over 15% of female biographies achieve Good, and nearly 15% Featured status, compared to just under 11% and 9% of male biographies, respectively. This suggests that, despite lower representation, female biographies seem to undergo more thorough development and refinement, as reflected now in both editing volume as well as article quality.

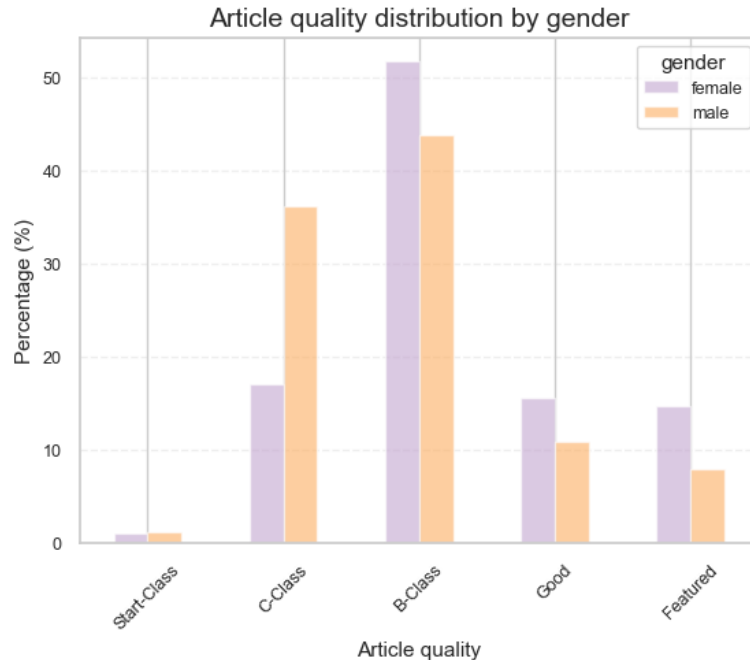


Figure 2: Distribution of article quality ratings by gender

3.2. Topic modeling and thematic analysis

Topic modeling revealed distinct thematic patterns in the biographies of men and women, reflecting differing narrative emphases. Across topics - as shown in *Table 2* - female biographies consistently and prominently feature words such as *woman*, *film*, *family*, *child*, and *mother*, suggesting a strong focus on personal relationships, possibly careers in film, as well as domestic roles. In contrast, topics among male biographies are dominated by terms like *war*, *state*, *work*, and *king*, indicating a heavier emphasis on political power and warfare, despite occasional appearances of words such as *family*, *father*, or *son*.

Table 2: LDA topic modeling results – most probable words in topics across genders

Topic #	Female biography topics	Male biography topics
1	family, woman, film, work, life, child	war, work, state, king, death, early
2	film, woman, mary, life, world, work	work, war, king, family, father, many
3	woman, child, life, mother, film, father	film, work, war, life, king, world
4	woman, work, life, film, child, war	work, war, state, early, life, world
5	woman, film, life, award, state, family	life, war, state, work, death, king
6	woman, film, life, family, work, child	war, work, king, state, made, life
7	film, woman, world, war, made, life	work, state, war, world, government, son
8	woman, film, life, death, queen, work	work, state, life, film, war, family

The accompanying word clouds (*Figure 3*) further illustrate these gendered discrepancies, with *film* and *woman* standing out for female biographies, while *war*, *work*, and *state* are central themes among men's pages. These results show gendered narratives in the content of biographies that are far from subtle, with professional contributions framed unevenly at best and unfairly at worst.



Figure 3: Word clouds of gendered biography topics

To try and quantify said discrepancies more thoroughly, the frequencies of relationship- and family-related words across biographies were computed. Figure 4 presents the density plot of those word counts by gender, where the distribution for males is much more concentrated between 0 and 25-50 words, indicating that most male biographies contain fewer family-related terms. In fact, the analysis reveals a striking contrast in the average occurrence of such terms, with female biographies averaging significantly higher at 42.57 (SD = 34.76), compared to 25.93 (SD = 21.78) in male biographies. The t-test result further confirms this disparity, with a t-statistic of -6.70 and a highly significant p-value ($p < 0.001$).

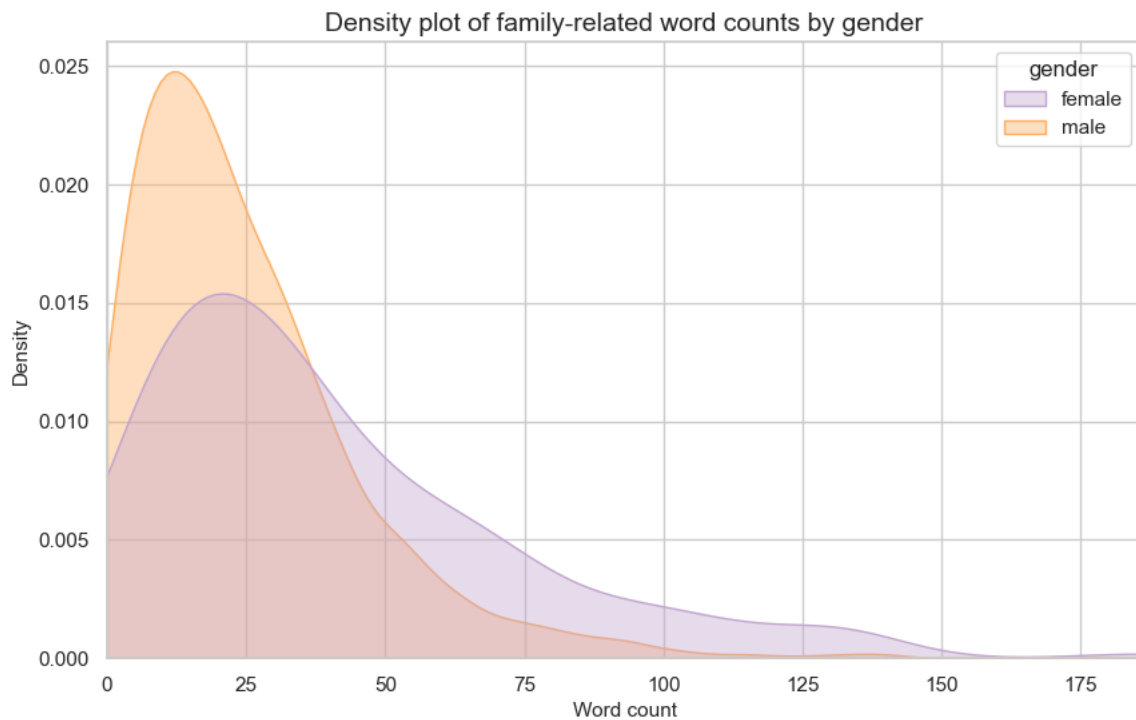


Figure 4: Gender-based distribution of family word count

4. DISCUSSION AND CONCLUSION

This study set out to explore gender disparities in Wikipedia’s Vital Articles list, analyzing both structural characteristics and thematic content. The findings paint a complex picture: while explicit structural inequalities in article formatting and quality ratings appear minimal, significant gender biases persist in representation and thematic focus.

Contrary to expectations, no substantial differences were found in structural page characteristics such as article length, number of sections, or number of hyperlinks. This suggests that once biographies reach the Vital Articles list, they are maintained with a level of consistency regardless of gender—possibly reflecting efforts to uphold Wikipedia’s standards

for these high-importance pages. However, this absence of structural disparities does not imply the absence of bias. The most glaring imbalance remains in representation: only about 10% of biographies in this dataset belong to women. This is particularly striking given that Wikipedia claims its Vital Articles list is free from biases (*Western* before the word *bias* in parentheses) (Wikipedia, n.d.). The underrepresentation of women at this fundamental level signals a deeper issue of who is deemed “notable” enough to be included in the first place, as it has been repeatedly pointed out by scholars (Falenska and Cetinoglu, 2021; Wagner et al. 2015). Further findings of this research on editorial attention and article quality showed that the two tend to reinforce each other, with female biographies receiving higher quality ratings as well as more editing activity compared to male biographies. However, despite significant results in the dataset without outliers, interpretation requires nuance. While higher levels of scrutiny for female biographies could be inferred from these findings, it is crucial to consider the entire dataset - especially in the context of this exploratory research - rather than relying solely on the filtered results. The “outliers” in edit activity - often representing the most contested or highly visible articles - are not to be discarded but should rather be seen as part of the broader Wikipedia ecosystem. Moreover, while structural inequalities in article formatting appear minimal, thematic analysis reveals deeply ingrained gendered narratives. Both topic modeling and word frequency analysis highlight clear differences in how male and female biographies are framed. Women’s biographies continue to feature more words related to family, personal relationships, and domestic roles, while men’s biographies are more centered on political, professional, and historical themes.

While this study provides valuable insights, future research could refine and expand the methodology to achieve even more rigorous results. For instance, within word frequency analysis, separating words related to gender, family, and personal relationships like the approach in Wagner et al. (2015) would allow for more nuanced interpretation of thematic biases. Further, expanding article quality metrics by including formerly featured articles and tracking changes over time (e.g. the proportion of female biographies that achieve higher quality ratings) would provide insight into whether disparities have narrowed over time. Finally, analyzing the revision history of the Vital Articles page itself could provide intriguing topic for future research, by examining gender dynamics in article selection. While a preliminary look into this aspect was conducted, methodological challenges made it difficult to integrate into this study.

To conclude, it needs to be emphasized that findings of this study have broader implications, as the gendered thematic disparities persist even at the highest editorial levels. If the ways in which women are portrayed still reflect traditional gendered expectations, the notion that their identities are more closely tied to personal relationships than professional achievements are reinforced. Such patterns raise questions about the ways in which knowledge is produced and reproduced.

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